

Enterprise VLAN & Trunking Lab

Phase 3 Report – VLAN Creation and Access Port Configuration

Project Information

Project Name: Enterprise VLAN & Trunking Lab

Phase: Phase 3 – VLAN Creation and Access Port Configuration

Simulation Software: Cisco Packet Tracer

Devices Configured: SW1, SW2

Status: Completed

1. Introduction

Following the successful deployment of the baseline switch configuration in Phase 2, the next stage of the project focused on implementing Virtual Local Area Networks (VLANs). VLAN technology enables the logical segmentation of a switched network by grouping devices into separate broadcast domains, regardless of their physical location.

During this phase, departmental VLANs were created on both Cisco switches, descriptive VLAN names were assigned, and access ports were configured to associate end devices with their corresponding VLANs. This logical segmentation establishes the foundation for improved network security, reduced broadcast traffic, and simplified administration.

At the conclusion of this phase, all end devices were successfully assigned to their designated VLANs. Inter-switch communication through trunking was intentionally postponed to Phase 4.

2. Objectives

The objectives of this phase were to:

- Create departmental VLANs on both switches.
- Assign descriptive names to each VLAN.
- Configure access ports for end-user devices.
- Associate switch interfaces with the correct VLAN.
- Verify VLAN membership using Cisco IOS commands.

- Validate communication between devices belonging to the same VLAN on the same switch.
 - Confirm isolation between different VLANs.
 - Prepare the network for IEEE 802.1Q trunk implementation.
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3. VLAN Design

The enterprise network was segmented into five VLANs.

VLAN ID	VLAN Name	Department	Purpose
10	MANAGEMENT	Management	Executive users
20	HR	Human Resources	Human Resources Department
30	SALES	Sales	Sales Department
40	IT	Information Technology	IT Department
99	NATIVE	Native VLAN	Trunk Native VLAN

Each VLAN represents an independent Layer 2 broadcast domain.

4. VLAN Implementation

The VLAN database was created on both SW1 and SW2 using Cisco IOS.

The following VLANs were successfully configured:

- VLAN 10 – MANAGEMENT
- VLAN 20 – HR
- VLAN 30 – SALES
- VLAN 40 – IT
- VLAN 99 – NATIVE

Maintaining identical VLAN databases across interconnected switches simplifies future expansion and supports consistent VLAN management.

5. Access Port Assignment

SW1

Interface	Connected Device	VLAN
FastEthernet0/1	MGMT-PC1	10
FastEthernet0/2	MGMT-PC2	10
FastEthernet0/3	HR-PC1	20
FastEthernet0/4	HR-PC2	20

SW2

Interface	Connected Device	VLAN
FastEthernet0/1	SALES-PC1	30
FastEthernet0/2	SALES-PC2	30
FastEthernet0/3	IT-PC1	40
FastEthernet0/4	IT-PC2	40

Each interface was explicitly configured as an access port using the `switchport mode access` command before being assigned to the appropriate VLAN. This ensures that each interface carries traffic for a single VLAN and prevents dynamic trunk negotiation.

6. Verification

Several Cisco IOS commands were used to verify the correctness of the configuration.

VLAN Database Verification

The following command was executed:

```
show vlan brief
```

Verification confirmed:

- VLANs 10, 20, 30, 40, and 99 were successfully created.
- VLAN names matched the network design.
- Access interfaces appeared under the correct VLAN.

Interface Verification

The following command was used to inspect individual interfaces:

```
show interfaces <interface> switchport
```

Verification confirmed:

- Administrative Mode: Static Access
- Operational Mode: Static Access
- Correct Access VLAN assignment

Configuration Verification

The running configuration was reviewed using:

```
show running-config
```

The review confirmed that:

- Access mode was configured correctly.
- VLAN assignments were applied to the intended interfaces.
- Interface descriptions matched the connected departments.

7. Connectivity Testing

Connectivity tests were performed using ICMP (ping).

Successful Communication

Communication was successfully established between devices belonging to the same VLAN and connected to the same switch.

Examples:

- MGMT-PC1 ↔ MGMT-PC2
- HR-PC1 ↔ HR-PC2
- SALES-PC1 ↔ SALES-PC2
- IT-PC1 ↔ IT-PC2

These successful tests confirmed that Layer 2 forwarding within each VLAN was functioning correctly.

VLAN Isolation

Communication between different VLANs was unsuccessful.

Examples:

- Management → HR
- Management → Sales
- HR → IT
- Sales → IT

These failed communication attempts demonstrate that VLANs successfully isolate broadcast domains and prevent direct Layer 2 communication between separate logical networks.

No inter-VLAN routing device was present during this phase; therefore, communication between VLANs was expected to fail.

8. Technical Concepts Applied

This phase reinforced the following networking concepts:

- Virtual Local Area Networks (VLANs)
 - Broadcast Domain Segmentation
 - IEEE 802.1Q VLAN Architecture
 - Access Ports
 - Static VLAN Assignment
 - Layer 2 Switching
 - MAC Address Learning
 - Cisco IOS VLAN Management
 - Enterprise Network Segmentation
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9. Best Practices Applied

The following industry best practices were implemented:

- Standardized VLAN numbering.
- Meaningful VLAN names.
- Explicit configuration of access ports.
- Interface descriptions documenting connected devices.

- Consistent VLAN databases across switches.
- Verification after each configuration stage.
- Saving the running configuration to startup memory.

These practices improve maintainability, scalability, and operational consistency in enterprise networks.

10. Deliverables

The following deliverables were completed during Phase 3:

- Departmental VLANs created on SW1.
 - Departmental VLANs created on SW2.
 - Native VLAN configured.
 - Access ports assigned to the appropriate VLANs.
 - Interface descriptions applied.
 - VLAN database verified.
 - Layer 2 connectivity validated within VLANs.
 - VLAN isolation verified.
 - Configuration saved to startup memory.
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11. Skills Developed

Upon completion of this phase, the following practical skills were acquired:

- Creating VLANs using Cisco IOS
- Managing VLAN databases
- Configuring access ports
- Assigning interfaces to VLANs
- Verifying switchport configurations
- Validating Layer 2 connectivity
- Troubleshooting VLAN membership
- Understanding enterprise LAN segmentation
- Applying network documentation standards

These competencies are fundamental for Network Engineers responsible for enterprise campus switching environments.

12. Lessons Learned

Several important networking principles were demonstrated during this phase:

- VLANs create logical networks independent of physical topology.
- Devices in the same VLAN communicate normally without requiring a router.
- Devices in different VLANs remain isolated until inter-VLAN routing is introduced.
- Proper interface configuration is essential for correct VLAN membership.
- Verification commands should always be used after configuration to confirm expected behavior.

These concepts form the basis for advanced enterprise switching technologies.

13. Preparation for Phase 4

Although identical VLAN databases now exist on both switches, the inter-switch connection is still operating as a default access link.

The next phase will convert this link into an IEEE 802.1Q trunk, allowing multiple VLANs to traverse a single physical connection. The trunk configuration will include:

- Trunk port configuration
- IEEE 802.1Q encapsulation (where applicable)
- Native VLAN configuration
- Allowed VLAN list
- Trunk verification using Cisco IOS commands

This enhancement will complete the Layer 2 infrastructure required for enterprise VLAN communication across multiple switches.

14. Conclusion

Phase 3 successfully implemented logical network segmentation through the creation of departmental VLANs and the configuration of access ports on both Cisco switches. Each department now operates within its own isolated broadcast domain, providing improved security, reduced broadcast traffic, and a scalable network architecture.

The successful verification of VLAN membership and intra-VLAN connectivity confirms that the Layer 2 switching infrastructure is functioning as intended. With the VLAN architecture now established, the network is fully prepared for the implementation of IEEE 802.1Q trunking in the next phase, enabling VLAN traffic to traverse the inter-switch link.